

Mini-Engagement Plan

SLM for Agriculture and Climate — agriculture-climate-slm-challenge

1. Selected Stakeholder Groups

For this project, we will focus on **agricultural extension worker associations** and **women smallholder farmer cooperatives** — the two groups most directly affected by, and most able to correct, the model's current gaps in field validation and data representation.

2. What Matters Most to Each Group

Extension worker networks care most about tools that extend their reach without undermining their authority — a model reflecting locally validated practice, one they can verify or correct rather than being silently overridden by. Women's cooperatives prioritize safe, direct access to agronomic information, since many depend on male relatives or infrequent extension visits for the same knowledge. They also want their specific labor, livestock, kitchen gardens, post-harvest work, represented rather than assumed generically. Both groups want a system that is accurate, verifiable, and respectful of who actually does the work.

3. Potential Harms and Benefits

Harms: extension workers may feel deskilled if the model is framed as a replacement rather than support, and farmer trust could erode if synthetic-data grounding produces confidently wrong advice. Women's cooperatives risk exclusion from data creation (the corpus wasn't co-created with them), misrepresentation of women-led practices if underrepresented in the 24 factsheets, and privacy issues if shared household phones limit private access. Benefits: extension workers gain a first-line answer tool reducing repetitive questions and freeing time for complex cases; women's cooperatives gain independent agronomic guidance, supporting economic participation without requiring male intermediaries.

4. Engagement Methods (via Action Catalogue)

Based on the Action Catalogue, **Participatory Design** with extension worker associations and **Distributed Dialogue** with women's farmer cooperatives are the strongest fits — the former leveraging extension workers' existing organizational structure and domain authority, the latter accommodating the geographic dispersion and access barriers that a centralized workshop would reproduce.

5. Why Each Method Fits

Participatory Design — Agricultural Extension Worker Associations Participatory design fits extension workers because their core concern is authority and accuracy, not merely access; they are the group best positioned to catch a specific failure mode already visible in this project: the current dataset is entirely synthetic, generated without input from anyone who actually delivers agricultural

advice in the field. A participatory design process directly closes that gap by treating extension workers as co-designers of the model's content and behavior, not as end users evaluating a finished product after the fact.

Concretely, this would take the form of structured co-design workshops run in two phases. In the first phase, extension workers would review the existing 24 factsheets and 45 training question-answer pairs against their own field experience, flagging where the synthetic content diverges from locally validated practice, where terminology doesn't match what farmers actually say, or where a factsheet omits a caveat they consider important (for example, a fertilizer rate that assumes soil conditions not universal across their service area). This phase treats extension workers' domain expertise as a correction mechanism for the dataset itself, not just feedback on the interface.

In the second phase, extension workers would review live model outputs generated during a pilot period, using their own real farmer questions rather than the benchmark's fixed test set. Because they are the ones fielding these questions daily, they can judge not just whether an answer is factually defensible, but whether it is the answer a farmer in their specific service area actually needs, something a generic accuracy check cannot capture. Their corrections and annotations would then feed back into a refined training set, closing the loop between deployment and data quality.

This method also directly addresses the authority concern: by giving extension workers editorial input into what the model says, participatory design frames the tool as an extension of their own expertise rather than a parallel, uncontrolled source of advice that could contradict them in front of farmers. That framing matters practically, if a farmer receives an answer from the model that conflicts with what their extension officer told them, trust in both the tool and the officer can erode. Involving extension workers early reduces the likelihood of that conflict arising in the first place, and gives them a legitimate channel to correct it when it does.

Participatory design also fits the practical constraints of this stakeholder group: extension workers are typically organized into associations or regional networks already, making structured, recurring workshops feasible to schedule without requiring new organizing infrastructure. Their existing professional incentive, reducing repetitive workload while preserving their role, aligns naturally with a method that gives them direct, ongoing influence over the tool rather than a one-time consultation.

Distributed Dialogue — Women Smallholder Farmer Cooperatives Distributed dialogue fits women's farmer cooperatives better than a centralized workshop model because of both their geographic dispersion and the access barriers the problem statement already identifies: many rural women face limited mobility, limited formal education access, and exclusion from spaces where agricultural decisions are formally discussed. A single centralized participatory design workshop, of the kind well-suited to extension worker associations, would risk reproducing exactly the exclusion this engagement is meant to address; requiring travel, time away from household and farm labor, and comfort speaking in a mixed or male-dominated setting, all of which disproportionately burden women.

Distributed dialogue instead allows multiple smaller, parallel conversations to happen across different cooperative locations, facilitated locally rather than requiring participants to converge on one site. This

matters concretely for this project: since the current dataset's 24 factsheets were generated synthetically and reflect only 7 crop categories and 3 agro-zones, there is a real risk that labor and knowledge specific to women, for instance, kitchen gardens, small livestock care, or post-harvest handling done predominantly by women in many of these communities, is underrepresented or generically framed. Distributed dialogue sessions, run through existing cooperative meeting structures women already attend, can surface these gaps directly from the people most affected by them, without asking women to adapt to an unfamiliar workshop format or travel burden first.

The dialogue format is also better suited than a single deliberative workshop for handling sensitive framing around privacy and autonomy. Because these conversations happen in smaller, familiar groups rather than one larger mixed forum, women are more likely to speak candidly about concerns such as shared-phone access limiting private use of the tool, or hesitation to ask questions that might be seen as challenging a husband's or elder's agricultural authority. A distributed format lets facilitators adapt each session's tone and language to the specific cooperative's context, rather than imposing one uniform engagement script across groups with different literacy levels, dialects, or comfort with technology.

Practically, findings from each distributed session would be aggregated centrally by the project team afterward, preserving the benefit of broad input while avoiding the exclusionary cost of requiring in-person centralization. This also creates a natural, lower-pressure entry point for cooperatives that might be wary of a formal research engagement; conversations can begin within a structure they already trust, rather than asking them to opt into an unfamiliar external process from the outset. Given that the project's current data has no women-specific input at all, this method offers the most realistic path to correcting that gap without adding new burdens that would deter participation.

6. Influence on Dataset and Model Design

Extension worker participatory design would most directly refine model outputs and add locally validated corrections to the training set, catching cases where synthetic content diverges from field practice or omits necessary caveats. Distributed dialogue with women's cooperatives would primarily reshape data sources and coverage; surfacing underrepresented practices (livestock, post-harvest, kitchen gardens) for inclusion in future factsheets, and informing consent and access design, such as ensuring the tool doesn't assume unshared private phone access. Together, both methods shift the dataset from a synthetic, single-perspective corpus toward one grounded in the people the model is meant to serve.